Features of Organisms Continued

Exam code: 0610 & 0970

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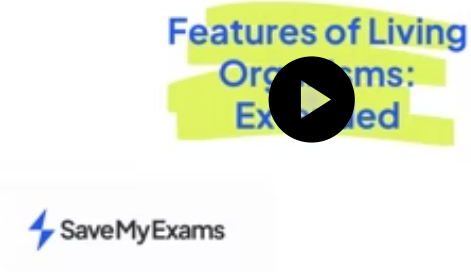


Did this video help you? Yes No

The Five Kingdoms: Extended

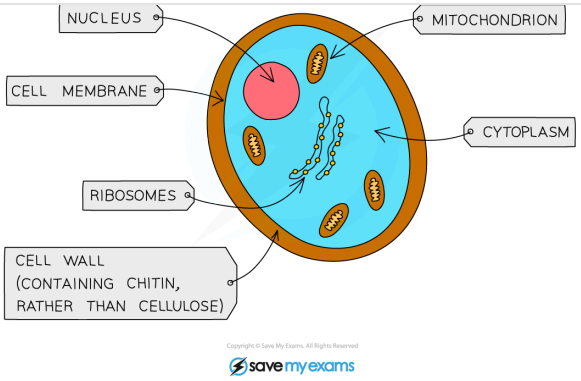
Extended Tier Only

- The first division of living things in the classification system is to put them into one of **five kingdoms**
- They are:
  - Animals
  - Plants
  - Fungi
  - Protoctists
  - Prokaryotes

- **Main features of all fungi (e.g. moulds, mushrooms, yeast)**
  - usually **multicellular**
  - cells have **nuclei** and **cell walls** not made from cellulose
  - do not photosynthesize but **feed by saprophytic** (on dead or decaying material) **or parasitic** (on live material) **nutrition**

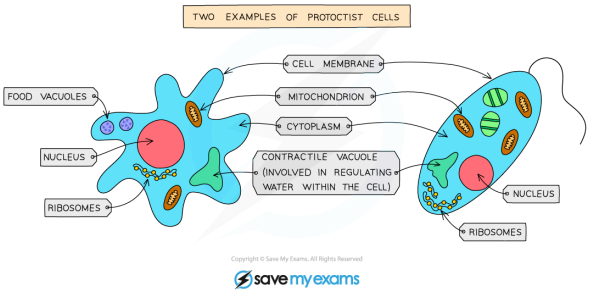
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



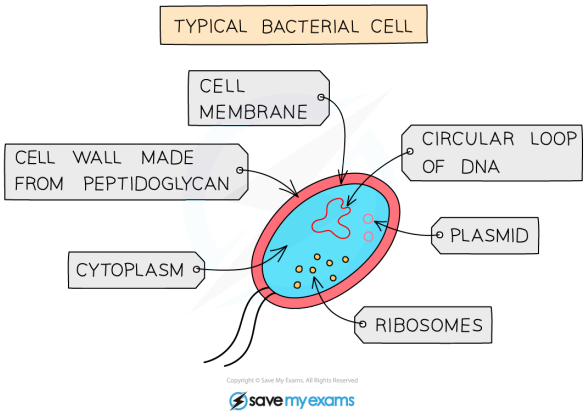
A typical fungal cell

- **Main features of all Protoctists (e.g. Amoeba, Paramecium, Plasmodium)**
  - **most are unicellular** but some are multicellular
  - all have a **nucleus**, some may have cell walls and chloroplasts
  - meaning **some protoctists photosynthesise** and **some feed on organic substances** made by other living things



Two examples of protoctist cells

- **Main features of all Prokaryotes (bacteria, blue-green algae)**
  - often **unicellular**
  - cells have **cell walls** (not made of cellulose) and **cytoplasm** but **no nucleus or mitochondria**



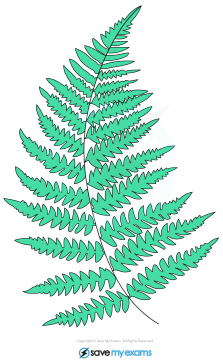
The Plant Kingdom: Extended

Extended Tier Only

- At least some parts of any plant are green, caused by the presence of the pigment **chlorophyll** which absorbs energy from sunlight for the process of **photosynthesis**
- The plant kingdom includes organisms such as **ferns and flowering plants**

Ferns

- Have leaves called **fronds**
- Do not produce flowers but instead **reproduce by spores** produced on the underside of fronds



Ferns

Ferns reproduce by spores found in the underside of their fronds

Flowering plants

- Reproduce sexually by means of **flowers and seeds**
- Seeds are produced inside the ovary found at the base of the flower
- Can be divided into two groups – **monocotyledons** and **dicotyledons**

Wheat plants are monocotyledons

Sunflowers are dicotyledons

How do you distinguish between monocotyledons and dicotyledons?

- **Flowers**
  - Flowers from **monocotyledons** contain petals in **multiples of 3** while flowers from dicotyledons contain petals in **multiples of 4 or 5**
- **Leaves**
  - Leaves from monocotyledons have **parallel leaf veins** while leaves from dicotyledons have **reticulated leaf veins**
    - Reticulated = form a web-like network throughout the leaf
  - Leaves from monocotyledons are **narrow** and grass-like while leaves from dicotyledons tend to have **broader** leaves that come in a wide range of shapes

Examiner Tips and Tricks

Identification of monocotyledons and dicotyledons comes up fairly frequently in the multiple choice paper and so it is worth learning the two differences between their flowers and leaves.

Viruses: Extended

Extended Tier Only

- Viruses are not part of any classification system as they are **not considered living things**
- They **do not carry out the seven life processes** for themselves, instead they **take over a host cell's metabolic pathways** in order to make multiple copies of themselves
- Virus structure is simply **genetic material** (RNA or DNA) inside a protein coat

Structure of a typical virus

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